UNIT 1 PRE-TEST

ANSWER KEY

# The Real Number System

*Answer each question, then check your work.*

Name _____ Date _____ Period _____

SCORE *3.0* Basic Understanding

## *1.1* Converting Rational Numbers

Convert each number as directed. Write fractions in simplest form.

**1.** Write as a decimal: 3/8

= **0.375**

e.g. 0.5

**Solution:**

3 ÷ 8 = 0.375. It terminates, so it is rational.

**2.** Write as a fraction: 0.25

= **1/4**

simplest form, e.g. 1/4

**Solution:**

0.25 = 25/100 = 1/4 in simplest form.

## *1.2* Real Number Classification

Classify the number as Rational (R) or Irrational (IR).

**3.** Classify: √144

Classification: **R**

type R or IR

**Solution:**

√144 = 12, a whole number, so it is rational.

## SCORE 4.0 Target Content

### 1.1 Converting Rational Numbers

Convert each number as directed. Write fractions in simplest form.

**4.** Write as a fraction: $0.\overline{6}$ (the 6 repeats)

= **2/3**

simplest form

**Solution:**

One digit repeats → put it over 9: 6/9 = 2/3.

**5.** Write as a decimal: $2\frac{5}{6}$ (2 and 5/6)

= **2.833...**

use ... for the repeating part

**Solution:**

2 5/6 = 2 + 5/6 = 2.8333... (the 3 repeats).

### 1.2 Real Number Classification

Classify the number as Rational (R) or Irrational (IR).

**6.** Classify: $\sqrt{20}$

`Classification:` **IR**

type R or IR

**Solution:**

20 is not a perfect square, so √20 = 2√5 ≈ 4.47... — non-terminating and non-repeating, so irrational.

*Simplify each expression first,* then classify the result.

**7.** Classify: $\sqrt{3} \times \sqrt{27}$

`Classification:` **R**

simplify, then R or IR

**Solution:**

√3 × √27 = √81 = 9, a whole number, so rational.

**8.** Classify: $\sqrt{32} \div \sqrt{2}$

Classification: **R**

simplify, then R or IR

**Solution:**

$\sqrt{32} \div \sqrt{2} = \sqrt{16} = 4$, a whole number, so rational.

### *1.3* Estimating Non-Perfect Square Roots

Estimate each value to the *nearest tenth* by clicking the number line.

**9.** Estimate: $\sqrt{40}$




**Solution:**

$6^2 = 36$ and $7^2 = 49$, so $\sqrt{40} \approx 6.32$, which rounds to 6.3.

**10.** Estimate: $\sqrt{95}$




**Solution:**

$9^2 = 81$ and $10^2 = 100$, so $\sqrt{95} \approx 9.747...$, which rounds to 9.7.

## SCORE *5.0* Advanced Content

### *1.2* Real Number Classification

Evaluate or simplify each expression if possible, then classify as Rational (R) or Irrational (IR).

**11.** Classify: $-(\sqrt{49} - \sqrt{36})$

Classification: **R**

simplify, then R or IR

**Solution:**

$-(\sqrt{49} - \sqrt{36}) = -(7 - 6) = -1$, an integer, so rational.

**12.** Classify: $3 + \sqrt{5}$

Classification: **IR**

type R or IR

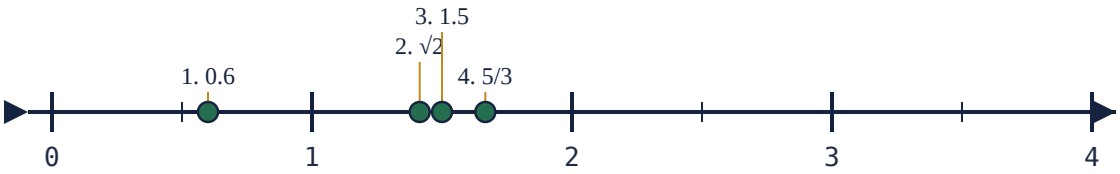
**Solution:**

√5 is irrational, and a rational number (3) plus an irrational number is irrational.

## *1.4* Ordering Real Numbers

Click the values in order from *least to greatest*. Each one drops onto the number line.

**13.** Set: { 0.6, √2, 1.5, 5/3 }



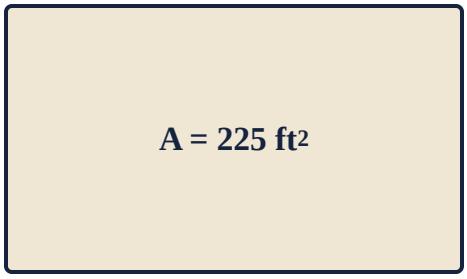

Order: **0.6, √2, 1.5, 5/3**

**Solution:**

As decimals: 0.6, √2 ≈ 1.41, 1.5, 5/3 ≈ 1.67 → least to greatest: 0.6, √2, 1.5, 5/3.

## *1.5* Area

A square garden has an area of 225 ft². Find the side length and the perimeter.




**14.** Find the side length and perimeter of the square garden.

Side length = **15 ft**

Perimeter = **60 ft**

**Solution:**

Side = √225 = 15 ft. Perimeter = 4 × 15 = 60 ft.